

Spatial Pressure Gravitational Theory — Flagship Manifesto & Quantum Foundation Milestone

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framework

Compiled in accordance with IPSTP v1.0 engineering practice and legal
interpretation specifications

Layer A + Layer B Public Document

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Project Website : <https://SGT.ac.cn>

Classification : Public (Layer A Fact Disclosure · IPSTP Rights Confirmation Carrier)

Abstract

This document serves as the flagship manifesto and milestone snapshot of the Spatial Pressure Gravitational Theory (SGT), declaring SGT's formal entry into the foundations of quantum mechanics. SGT posits that the vacuum is filled by a continuous elastic medium—"Spatial Energy"—with matter as topological defects in this medium, and gravity arising from the tension gradients produced when matter

stretches the medium. Through systematic theoretical construction and multiple rounds of independent audits, SGT has achieved full closure of its classical field background—with independent numerical solutions for the Ω -A1 ODE produced, the medium-metric coupling constant rigorously derived from action variation, and five constitutional anchors verified and corrected with high precision. In the domain of quantum foundations, SGT provides **A-level derivations** for spin- $\frac{1}{2}$, identical particles, and antiparticles that are completely independent of quantum mechanical axioms—using no quantum parameters or postulates. SGT supplies a deterministic ontology for quantum mechanics: the wavefunction is the slowly-varying envelope of the medium's strain field, and entanglement is the non-separability of global PDE solutions in a classical elastic medium. This paper follows the IPSTP protocol's three-part rights confirmation structure of **Conditions** → **Constraints** → **Results**, strictly distinguishes proven theorems from model predictions, and publicly releases for the first time the evidence tier system for SGT's quantum docking.

Keywords : Spatial Pressure Gravitational Theory; IPSTP rights confirmation; foundations of quantum mechanics; topological defects; medium elasticity; topological origin of spin- $\frac{1}{2}$; bridge constant; deterministic ontology

Prefatory Rights Confirmation Information

Item	Content
Sole Author	Li, Zhijun

ORCID	0009-0004-8456-7107
Constitutional Basis	V4.2.2
IPSTP Preservation Level	Tier 3: Maximum Preservation (Spatial + Temporal Anchoring)
IPSTP Protocol Application Guide	10.5281/zenodo.20557825
Zenodo Archive DOI	10.5281/zenodo.20602406
TSA Timestamp Certification	TSA-01-20260609727334209
IPFS CID	QmVaSLPHd11YC5eRkGgJRadeijau5BRtsGmSrBXUBDf1XC
DCI Digital Copyright Identifier	RCCS00ANT.202606023452993300
Filing Channels	TSA Trusted Timestamp · DCI Digital Copyright · National Copyright Center · Academic Preprints · AntChain Blockchain · Digital Copyright Platform · Depository Public Chains
Open Source Repository	github.com/letiangogo/SGT
Project Website	sgt.ac.cn

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Note : IPSTP v1.0 is a custom intellectual property spatiotemporal proof protocol of this project, used for engineering-oriented rights confirmation. ZKAV is a custom zero-knowledge academic verification protocol of this project. Neither constitutes an international or industry standard; their interpretation and revision rights belong to the SGT project team.

Verification Layer Lock Information

Item	Content
Verification Layer File	SGT-Verification-Layer-Lock-File-v2.0.rar
SHA-256	4a354a8456cad88d4f6ff405dd27e135530a9161b9120b2c02179e1a44bd3c1
TSA Certificate No.	TSA-01-20260609727334209
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Four-Code Unity Rights Confirmation Statement : The SHA-256 hash, TSA timestamp certification code, Zenodo DOI archive number, and IPFS Content Identifier (CID) of this document form an integrated, immutable spatiotemporal evidence chain. Any third party may independently verify the integrity and temporal priority of this document through these four codes.

Declaration

This document is the flagship manifesto and milestone snapshot of the Spatial Pressure Gravitational Theory, compiled according to the IPSTP v1.0 three-layer stripping structure. The core principle is the separation of verification rights from access rights.

Layer A (Conditions and Conclusions Layer) publicly discloses all theoretical conditions and precise numerical conclusions of SGT; **Layer B (Constraints Framework Layer)** publicly discloses all theoretical constraints and structural rules of SGT. The above information is sufficient to enable researchers with relevant numerical backgrounds to independently verify the conclusions through independent engineering paths, without requiring access to Layer C.

Layer C (Implementation Source Code Layer) is retained as a trade secret in accordance with IPSTP v1.0. Layer C shall only be directionally retrieved under legally prescribed circumstances, such as judicial evidence collection, administrative investigation, or written authorization by the author.

This document constitutes a Verifiable Disclosure Event (VDE). Should identical or

highly similar mappings appear subsequently, independent verification can be conducted through the spatiotemporal anchoring records.

1. Theoretical Project

Theory Name : Spatial Pressure Gravitational Theory (SGT)

Theoretical Positioning : An effective theory of gravity with a vacuum continuous elastic medium as its ontology. Matter consists of topological defects in the medium; gravity is the tension gradient produced when matter stretches the medium. SGT does not modify the mathematical formalism of General Relativity, but provides it with an elastic-medium ontological interpretation. It covers all gravitational scenarios, from galactic rotation curves to black hole horizons, from the Big Bang to the foundations of quantum mechanics.

Theoretical Sectors : Foundations | Lagrangian Field Theory | Spherically Symmetric Black Holes | Rotating Black Holes | Primordial Universe and Singularity Elimination | Cosmological Linear Perturbations and CMB | Compact Binary Dynamics | Gravitational Structure Stability | Testable Observational Predictions | Foundations of Quantum Mechanics

2. Conditions Set P (Attributed to Layer A)

2.1 Axiom System

No.	Axiom	Content
SGT-P001	A1	The spatial energy medium exists. The vacuum is filled by a continuous elastic medium. The medium is the physical ontology of SGT; the spacetime metric is the mathematical translation layer of the medium's stretching state.
SGT-P002	A2	Matter consists of localized defects in the medium, stretching the medium outward when occupying space. Particle-medium coupling constant $\kappa_{\text{particle}} = G/2$; medium-metric coupling constant $\kappa_{\text{medium}} = 8\pi GK$. The two are conceptually distinct.
SGT-P003	A3	Gravity is the tension gradient of the medium. A uniform tension produces no gravity; a gradient in tension produces a net force directed toward the center of mass. Gravitational waves are elastic wave escape phenomena—when the source shakes violently and the medium cannot keep up, the strain gradient detaches from the source region and propagates outward at the medium's intrinsic signal speed c .

2.2 Core Variables and Field Equations

No.	Condition	Content
SGT-P004	Stretch degree	$f = 1 - A_0 \in [0,1]$
SGT-P005	Order parameter constitutive law	$\chi = (1-f)/(1+f^2), \chi \in [0,1]$

SGT-P006	Effective gravitational constant	$G_{\text{eff}} = G \cdot \chi(f)$
SGT-P007	Static field equation form	Under spherical vacuum symmetry, reduces to the Ω -A1 ODE system: $dm/dr = (\kappa_{\text{medium}}/2)r^2[U+2U'(1-f)]$, $dv_0/dr = [m-(\kappa_{\text{medium}}/2)Ur^3]/[r(r-2m)]$
SGT-P008	Modified Friedmann equation	$H^2 = (8\pi G/3)\chi\rho_m + (8\pi GK/3)\rho_T(f)$, derived from action variation
SGT-P009	f kinetic equation	$\ddot{f} + 3H\dot{f} + V_{\text{eff}}'(f) = 0$, derived from action variation

2.3 Parameter Declaration

No.	Condition	Content
SGT-P010	Number of independent parameters	Three—observationally constrained parameters K and f_c , plus the phenomenological theoretical parameter n (medium potential energy accumulation index). Δf is determined by f_c , κ_{medium} by K , and f_{eq} by n ; all are non-independent parameters.
SGT-P011	Fundamental constants	G (Newtonian gravitational constant), c (speed of light), $\hbar$ (topological-quantum bridge constant—classical topology provides quantum structure, $\hbar$ provides dimensional scale. The numerical value of $\hbar$ currently relies on experimental measurement; its independent derivation is the Q2-stage objective).

2.4 Lagrangian Field Theory Framework

No.	Condition	Content
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SGT-P012	Total action structure	$S = \int [R/(16\pi G) - \rho_{\text{medium}}] \sqrt{-g} d^4x + S_{\text{matter}}$; the medium's elastic energy couples to the metric only in proportion K.
SGT-P013	Kinetic term	The kinetic term $F = 1$ holds exactly in the Lagrangian description (Grade A closure).
SGT-P014	Stress constitutive relations	Derived from the first variation of the elastic strain energy; fully explicit and closed.

3. Constraints Set B (Attributed to Layer B)

3.1 Non-emptiability and Singularity Elimination

No.	Constraint	Proof Strength
SGT-B001	$\chi \geq 0$ always holds; the medium cannot be completely emptied.	Grade A multi-dimensional proof
SGT-B002	Equivalent proposition: $C_{rr} \rightarrow \infty \iff \chi \geq 0$	Grade A
SGT-B003	$f \leq 1$ locked at the action level	Grade A
SGT-B004	Cosmic Censorship hypothesis: Non-emptiability prohibits naked singularities.	Theorem-level corollary

3.2 Elastic Constant System

No.	Constraint	Proof Strength
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SGT-B005	Medium physical state: Prestressed transversely isotropic elastic solid.	Grade A
SGT-B006	Anisotropic stress fingerprint: $P_\theta < P_r$ holds globally.	Grade A
SGT-B007	Elastic constant system closure: 6 constants (4 independent + 2 constrained).	Grade A
SGT-B008	As $\chi \rightarrow 0$: $C_{rr} \rightarrow \infty$, $C_{\theta\theta} \rightarrow 0$, $C_{44} \rightarrow 0$, $C_{66} \rightarrow 0$.	Grade A cross-validated
SGT-B009	χ as order parameter: organizes elastic phase transitions.	Grade A
SGT-B010	Frozen layer structure: five layers; freezing front $\approx$ photon sphere.	Grade B+
SGT-B011	Solitary waves do not exist.	Grade A
SGT-B012	Physical distinction between W (deformation work) and U (stored potential energy): In the strong-field region, $W \rightarrow 0$ while $U \rightarrow K/2$; the two decouple.	Confirmed

3.3 Causal Structure and Stability

No.	Constraint	Proof Strength
SGT-B013	Causal structure theorem: The inner shell is causally closed.	Grade A
SGT-B014	Global hyperbolicity.	Grade A
SGT-	Energy conservation: $dE/dt = 0$.	Grade A

B015		
SGT-B016	Nonlinear stability rigorously proven.	Grade A
SGT-B017	Onsager reciprocity holds exactly.	Grade A
SGT-B018	Static solution uniqueness theorem.	Grade A
SGT-B019	$\chi(f)$ qualitative equivalence theorem.	Grade A-
SGT-B020	Metric-medium separation principle.	Rigorously proven

3.4 Black Hole Physics

No.	Constraint	Proof Strength
SGT-B021	Curvature scalars remain finite at the absolute horizon.	Grade A
SGT-B022	Smarr formula holds exactly.	Grade A
SGT-B023	Area law holds strictly.	Grade A

3.5 Rotating Black Holes

No.	Constraint	Proof
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		Strength
SGT-B024	Local Effective Mass Approximation verified up to a ≤ 0.5 .	Grade A
SGT-B025	G_eff negative feedback mechanism.	Grade A

3.6 Cosmology

No.	Constraint	Proof Strength
SGT-B026	Singularity elimination theorem: $f(0)=1 \rightarrow H(0) < \infty$.	Grade A
SGT-B027	Equivalence theorem: $f(0)=1 \iff \chi(0)=0 \iff H(0) < \infty$.	Grade A
SGT-B028	Primordial phase structure: Frozen de Sitter $\rightarrow$ Thawing (Tachyon-driven) $\rightarrow$ Fast-roll $\rightarrow$ Normal expansion.	Qualitative closure
SGT-B029	Unified mechanism with black hole singularity elimination.	Grade A
SGT-B030	Standard slow-roll not viable (honest assessment).	Noted
SGT-B031	Frozen-phase scalar perturbation suppression: $\chi=0 \rightarrow G_{\text{eff}}=0$.	Grade A algebraic conclusion

3.7 Compact Binaries

No.	Constraint	Proof
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		Strength
SGT-B032	χ as an intrinsic scalar degree of freedom.	Grade A
SGT-B033	Scalar dipole radiation originates from the time variation of χ .	Qualitative

3.8 Gravitational Structure Stability

No.	Constraint	Proof Strength
SGT-B034	Three-tier rigidity map: Horizon ϵ^0 , ISCO ϵ^2 , Photon sphere ϵ^1 .	Grade A
SGT-B035	Safety margin: $71\times$.	Grade A
SGT-B036	All 8 CH-series tasks completed.	Grade A

3.9 Theoretical Self-Consistency

No.	Constraint	Proof Strength
SGT-B037	Full-chain consistency audit (TR-065): Six-layer penetration—axioms $\rightarrow$ constitutive $\rightarrow$ field equations $\rightarrow$ anchors $\rightarrow$ predictions $\rightarrow$ quantum docking—zero fractures.	Green pass
SGT-B038	Deep implicit loop audit (TR-067): No $A \rightarrow B \rightarrow C \rightarrow A$ logical loops detected.	Green pass

SGT-B039	$\kappa_{\text{medium}} = 8\pi G K$ rigorously derived from action variation (TR-063).	Derivation-grade
SGT-B040	$\kappa_{\text{particle}} = G/2$ independently confirmed by the equivalence principle (TR-040), independent of κ_{medium} (verified by TR-066 T7).	Derivation-grade

4. Conclusions Set Y (Attributed to Layer A)

4.1 Classical Field Background

No.	Conclusion	SGT Value	GR Value	Proof Strength
SGT-Y001	Horizon radius	1.66M	2M	Grade A
SGT-Y002	ISCO radius (G_{eff} modulated)	4.18M	6.00M	Grade A
SGT-Y003	Photon sphere radius	$\sim 3.0M$	3M	Grade A
SGT-Y004	Hawking temperature ratio	$0.628 \times T_H^{\text{GR}}$	T_H^{GR}	Grade A
SGT-Y005	Horizon entropy	$\eta = 2.3 = 1.0 + 1.3$	1.00	Grade B

4.2 Rotating Black Holes

No.	Conclusion	SGT Value	GR	Proof
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			Value	Strength
SGT-Y006	ISCO rotation coefficient C_1	≈ -0.47 (to be recalculated with new background)	-3.266	Grade A
SGT-Y007	Observable difference in slow-rotation regime	$< 0.3\%$	—	Grade A
SGT-Y008	Shadow D-shape difference	$< 0.05\%$	—	Grade A

4.3 Primordial Universe and CMB Fingerprints

No.	Conclusion	Value / Feature	Proof Strength
SGT-Y009	Initial Hubble constant $H(0)$	1.023×10^{-1}	Grade A
SGT-Y010	Initial total density	1.250×10^{-3}	Grade A
SGT-Y011	Scalar spectrum low- ℓ suppression	$\chi \rightarrow 0 \rightarrow G_{\text{eff}} \rightarrow 0$	Grade A
SGT-Y012	Early Ψ - Φ separation	Elastic anisotropy	Grade A
SGT-Y013	Super-horizon $\mathcal{R}$ non-conservation	$\delta P_{\text{Nad}} \neq 0$	Grade A
SGT-Y014	Tensor-to-scalar ratio r low- ℓ enhancement	P_{T} normal, P_{S} suppressed	Qualitative

SGT-Y015	Tachyon perturbation generation	$V_{\text{eff}}' < 0$ driven	Grade B+
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4.4 Testable Predictions

No.	Prediction	Value / Feature	Verification Instrument	Proof Strength
SGT-Y016	Fe K α red-wing shift	500–900 eV	Athena 2027+	Grade A
SGT-Y017	Gravitational wave echo time delay	$\sim 14.2\text{M}$	3G detectors	Grade B
SGT-Y018	Echo frequency dependence	$R(\omega) \propto \omega^\alpha$ ($\alpha > 0$)	3G detectors	Grade B
SGT-Y019	Neutron star TOV no correction	—	NICER verified	Grade A
SGT-Y020	Rotating black hole shadow	Difference < 0.3%	EHT verified	Grade A
SGT-Y021	CMB scalar spectrum low- ℓ suppression	$\ell < 30$	CMB-S4	Grade A
SGT-Y022	CMB early Ψ - Φ separation	E-mode	CMB-S4	Grade A
SGT-Y023	ISCO rotation suppression	$C_1 \approx -0.47$	LISA 2035+	Grade A
SGT-Y024	Primordial gravitational wave low-frequency	$G_{\text{eff}} \rightarrow 0$	3G detectors	Qualitative

	suppression			
SGT-Y025	Proton frozen core	Form factor deviation from dipole	PRad / MESA / EIC	Exploratory framework

4.5 Foundations of Quantum Mechanics (Locked at V4.2.2)

Grade A: Fully Independent of Quantum Mechanics (3 items)

No.	Conclusion	Core Argument	Physical Meaning
SGT-Y026	Spin- $\frac{1}{2}$	Z_2 disclination $\rightarrow 4\pi$ periodicity	Spin is not an intrinsic attribute; it is an inevitable consequence of classical topology.
SGT-Y027	Identical particles	Topological defect indistinguishability	Identity is not a postulate; it is topological equivalence.
SGT-Y028	Antiparticles	Topological charge ± 1 and annihilation	Antiparticles are not products of the Dirac equation; they are topological duals.

****Grade B+*: Structurally Independent, Bridge-scaled (5 items)****

No.	Conclusion	Structure Provided by SGT
SGT-Y029	Existence of $\hbar$	Adiabatic invariant concept
SGT-Y030	Uncertainty principle	Positioning limit of the defect core radius
SGT-	Bound states	V_{eff} potential well (purely SGT-derived)

Y031		
SGT-Y032	Entanglement mechanism	Elastic coupling → non-separability of global PDE solutions

Grade B- (1 item) : Schrödinger equation form—the equation form can be derived from KG equation + SVEA.

Grade C (1 item) : SGT-independent derivation of Bell inequality violation—pending Q4 phase completion.

Positioning of $\hbar$: $\hbar$ is introduced in SGT as a topological-quantum bridge constant—classical topology provides the quantum structure, and $\hbar$ provides the dimensional scale. Its methodological status is analogous to c linking space and time in Special Relativity, and k_B linking the microscopic and macroscopic in Statistical Mechanics.

5. Layer C Retention Statement

The core technical assets of SGT theory are retained as Layer C in accordance with IPSTP v1.0, kept permanently private by default, and fully protected under trade secret law throughout.

Layers A and B have provided sufficient information for third parties to reconstruct constraints and independently verify conclusions; Layer C is absolutely not a necessary condition for verification.

Legal Access Exceptions : Layer C shall be directionally decrypted and verified only under legally prescribed circumstances, such as judicial evidence collection, administrative investigation, or written authorization by the author.

6. Intellectual Property Protection System Statement

The intellectual property protection of the Spatial Pressure Gravitational Theory has constructed a multi-level, cross-platform, judicially-grade depository and verification system:

Rights Confirmation Protocol Layer : IPSTP v1.0 Intellectual Property Spatiotemporal Proof Protocol.

Spatiotemporal Anchoring Layer : SHA-256 hash solidification + TSA trusted timestamp + IPFS decentralized permanent storage.

Academic Depository Layer : Zenodo DOI permanent archive + academic preprint DOI registration.

Legal Filing Layer : National Copyright Center registration + DCI digital copyright identifier.

Technical Verification Layer : ZKAV zero-knowledge academic verification protocol.

Four-Code Unity : SHA-256 + TSA + DOI + IPFS CID form an integrated, immutable spatiotemporal evidence chain.

7. Directivity Statement

The 86 technical anchors disclosed in this document—14 Premises P, 40 Constraints B, and 32 Conclusions Y—together constitute the indivisible theoretical fingerprint of the Spatial Pressure Gravitational Theory. When any subsequent theory

simultaneously satisfies the following conditions, it shall be deemed to fall within the condition space already established by SGT: (a) its axiom system is substantially similar to SGT-P001 through SGT-P003 of this document; (b) its field equation framework and constitutive law forms are equivalent expressions of SGT-P004 through SGT-P009; (c) its numerical conclusions exhibit systematic consistency with SGT-Y001 through SGT-Y032 of this document.

8. Mandatory Standard Marking

This paper complies with the IPSTP v1.0 Tier-3 preservation mechanism, and has completed spatial and temporal anchoring. SHA-256:

4a354a8456cad88d4f6ff405dd27e135530a9161b9120b2c02179e1a44bd3c1. TSA:

TSA-01-20260609727334209. DOI: 10.5281/zenodo.20602406. IPFS:

QmVaSLPHd11YC5eRkGgJRadeijau5BRtsGmSrBXUBDf1XC. Four codes unified, the strongest rights confirmation.

9. Honest Marking and Conclusion

SGT is a candidate theory of gravity and has not yet completed full peer review.

Forty-five liabilities have been cleared, and twenty remain. The full-chain consistency audit has passed green; the deep audit has detected no implicit logical loops. The Grade A quantum docking achievements are topological derivations independent of QM.

With a unified medium-mechanics framework, SGT has provided answers in two

extreme regimes—black holes and cosmology—that General Relativity could not. A black hole is a frozen state of the medium; the Big Bang is a phase transition of medium stiffness exhaustion. Spin- $\frac{1}{2}$ is an inevitable consequence of classical topology; identical particles reflect topological equivalence; antiparticles are topological duals. These core features of quantum mechanics emerge from classical physics. General Relativity used spacetime curvature to explain *what* gravity is; SGT uses medium tension to explain *why* gravity is—while simultaneously providing classical topological roots for *why* quantum mechanics possesses its hallmark features. The SGT framework stands ready to be tested by the stars.

10. Anchor Statistics

Category	Count
Premises P (Layer A)	14
Constraints B (Layer B)	40
Conclusions Y (Layer A)	32
Total Anchors	86
Theoretical Sectors	10
IPSTP Preservation Level	Tier 3 (Maximum Preservation)

Four-Code Unity Rights Confirmation :

- SHA-256:

4a354a8456cad88d4f6ff405dd27e135530a9161b9120b2c02179e1a44bd3
c1

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**Spatial Pressure Gravitational Theory — Flagship Manifesto & Quantum
Foundation Milestone. Constitutional Basis V4.2.2. SGT formally enters the
foundations of quantum mechanics.**